Problem 3: Left and Right Sum Differences (easy)

Problem Statement

Given an input array of integers `nums`, find an integer array, let's call it `differenceArray,` of the same length as an input integer array.

Each element of `differenceArray`, i.e., `differenceArray[i],` should be calculated as follows: take the sum of all elements to the left of index `i` in array `nums` (denoted as `leftSum[i]`), and subtract it from the sum of all elements to the right of index `i` in array `nums` (denoted as `rightSum[i]`), taking the absolute value of the result. Formally:

$$\text{differenceArray}[i] = |\ \text{leftSum}[i] - \text{rightSum}[i]\ |$$

If there are no elements to the left/right of `i`, the corresponding sum should be taken as 0.

Examples

Example 1:

- **Input**: `[2, 5, 1, 6]`
- **Expected Output**: `[12, 5, 1, 8]`
- **Explanation**:
  - For i=0: |(0) - (5+1+6)| = |0 - 12| = 12
  - For i=1: |(2) - (1+6)| = |2 - 7| = 5
  - For i=2: |(2+5) - (6)| = |7 - 6| = 1
  - For i=3: |(2+5+1) - (0)| = |8 - 0| = 8

Example 2:

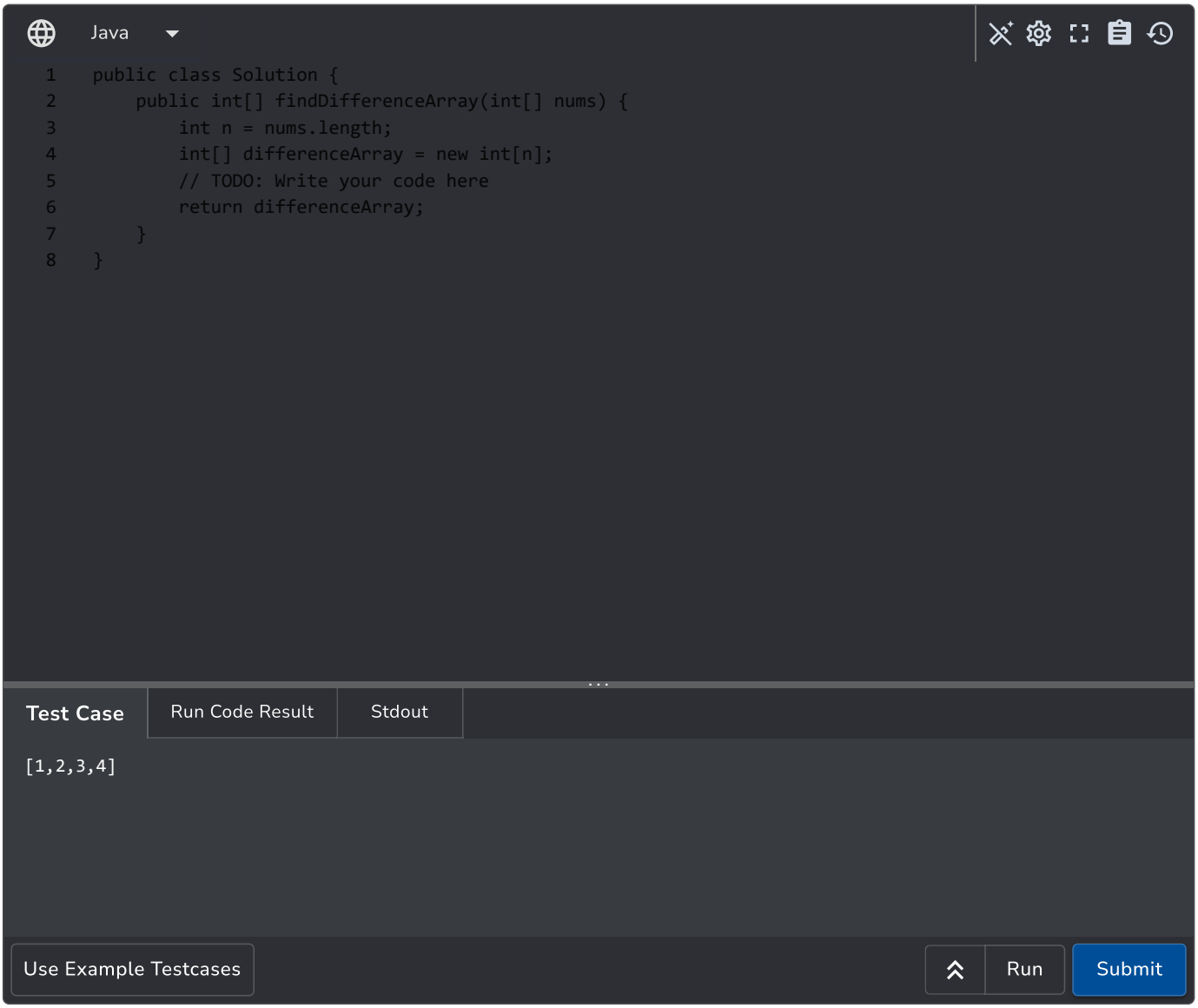
- **Input**: `[3, 3, 3]`
- **Expected Output**: `[6, 0, 6]`
- **Explanation**:
  - For i=0: |(0) - (3+3)| = 6
  - For i=1: |(3) - (3)| = 0
  - For i=2: |(3+3) - (0)| = 6

Example 3:

- **Input**: `[1, 2, 3, 4, 5]`
- **Expected Output**: `[14, 11, 6, 1, 10]`
- **Explanation**:
  - Calculations for each index i will follow the above-mentioned logic.

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